

9.3 Direct Preference Optimization

Same pairs. Implicit $r = \beta \log(\pi/\pi_{\text{ref}})$. $Z(x)$ cancels. Offline.

These notes match the lecture slides. Use **(slides)** on the course hub for the deck.

Direct Preference Optimization (DPO) (Rafailov et al., 2023) fits a policy to chosen/rejected pairs **without** training an RM and **without** PPO. The reward is written in closed form from the policy and a frozen reference, then the Bradley–Terry model becomes a classification loss on logits.

1. What DPO is

RLHF solves $\max_{\pi} \mathbb{E}[r] - \beta \text{KL}(\pi \parallel \pi_{\text{ref}})$ with a learned RM and an on-policy loop. **DPO** uses the same preference pairs and skips both the RM and PPO. Under Bradley–Terry, the optimal reward can be written from the policy and a frozen reference,

$$r(x, y) = \beta \log \frac{\pi(y \mid x)}{\pi_{\text{ref}}(y \mid x)} + \text{const}(x).$$

Plug that into the RM logistic loss and the r parameters disappear. You train π_{θ} directly, offline. The derived reward has a **partition** $Z(x)$ that cancels because Bradley–Terry only sees a difference. KTO and IPO sit in the same family; this hour is DPO.

If $\pi_{\theta} = \pi_{\text{ref}}$, both log-ratios are 0, the argument of σ is 0, and $\sigma(0) = 1/2$. Then $\mathcal{L}_{\text{DPO}} = \log 2 \approx 0.693$. That is the **start** of training, not a bug. Lab 9 will compute this on toy scalar logits, not on an LLM.

2. Why we use it

RLHF's PPO stage is heavy: sampling, a critic, a moving policy, stale rollouts. DPO is a classification loss on logits of y_w and y_l . You have pairs, a GPU, and no appetite for a PPO stack. Offline: the pair set is fixed, so you do not explore new y unless you collect more.

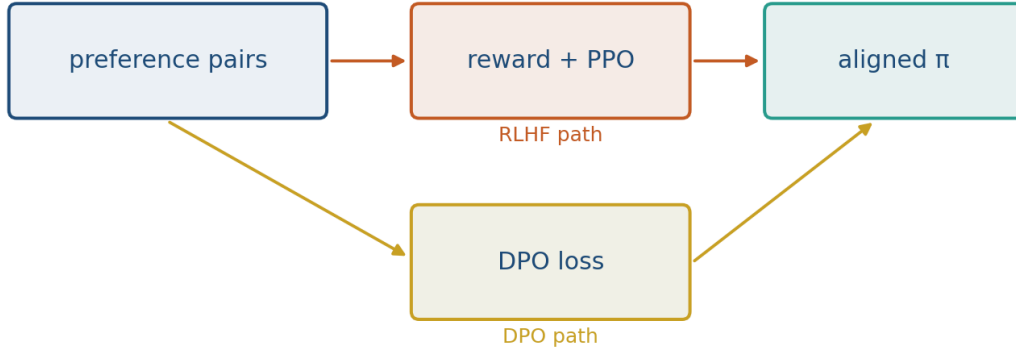
That is also the limit. If the SFT model never proposed good y_w , DPO cannot invent them from nothing. PPO can **sample new** y that were not in the pair file; offline DPO cannot. Implicit reward can still overfit the pair set. Report β , the reference checkpoint, and a held-out preference accuracy (how often π_{θ} assigns higher log-prob to y_w than to y_l). That number is not "the model is aligned"; it is "the model copied this rater."

3. Architecture

Two networks, one frozen: policy π_{θ} and reference π_{ref} (usually SFT). No critic. No sampling-in-the-loop required. You need log-probabilities of **both** completions under π_{θ} and under π_{ref} . The reference stays

frozen. Dropping π_{ref} because “there is no RM” removes the leash; the reference **is** the leash.

DPO fits the same pairwise data without an explicit reward model



The data path is the same pair file as note 9.1. The compute path is a forward on y_w and a forward on y_l for each of π_θ and π_{ref} (four logprob sums, or two if you cache the frozen reference). There is no RM head.

4. How it works, step by step

1. Start from SFT. Freeze that checkpoint as π_{ref} . Copy it as π_θ .
2. For each pair (x, y_w, y_l) , compute log-prob of both completions under both models.
3. Form the two log-ratios $\log(\pi_\theta / \pi_{\text{ref}})$ and subtract loser from winner. Scale by β .
4. Push that difference through σ and take $-\log$. Gradient raises relative mass on y_w versus y_l , compared with the reference.
5. Absolute log-prob can fall on both sequences; **relative** log-prob is what DPO scores. Write the argument of σ as $\beta(\Delta \log \pi_\theta - \Delta \log \pi_{\text{ref}})$ if it helps.

At init, $\pi_\theta = \pi_{\text{ref}}$ and the loss is $\log 2$. After a few steps the winner log-ratio should exceed the loser log-ratio. The frozen reference is still there: without it, π_θ can raise both log-probs or collapse. β is the same kind of leash as in RLHF: it scales how hard you push away from π_{ref} .

5. Mathematical formulas

Closed-form reward, then DPO loss:

$$r(x, y) = \beta \log \frac{\pi(y | x)}{\pi_{\text{ref}}(y | x)} + \text{const}(x),$$

$$\mathcal{L}_{\text{DPO}} = -\log \sigma \left(\beta \log \frac{\pi_\theta(y_w | x)}{\pi_{\text{ref}}(y_w | x)} - \beta \log \frac{\pi_\theta(y_l | x)}{\pi_{\text{ref}}(y_l | x)} \right).$$

If $\pi_\theta = \pi_{\text{ref}}$, both log-ratios vanish and $\mathcal{L}_{\text{DPO}} = \log 2$. After a few steps, toy log-probs (sums over tokens): $\log \pi_\theta(y_w) = -2.0$, $\log \pi_{\text{ref}}(y_w) = -2.5$, $\log \pi_\theta(y_l) = -3.0$, $\log \pi_{\text{ref}}(y_l) = -2.0$. Log-ratio winner: $-2.0 - (-2.5) = 0.5$. Log-ratio loser: $-3.0 - (-2.0) = -1.0$. Difference $0.5 - (-1.0) = 1.5$. $\sigma(1.5) \approx 0.818$, loss $-\log 0.818 \approx 0.201$.



Init loss $< \mathcal{L}_{\text{DPO}} = \log 2$

log2, then a 1.5 logit gap" src="graphics/9.3-dpo/dpo-init.png" />

6. Positive points and negative points

Positive.

- Offline classification loss: no RM head, no PPO loop, no on-policy sampling.
- Same pair file as the RM lecture; the closed-form r is why the RM can disappear.
- Init loss $\log 2$ is a checkable number (Lab 9).
- β is still an explicit leash toward π_{ref} .

Negative.

- Still needs pairs and a frozen reference. Dropping the reference is not a simplification.
- Can overfit the pair set. Held-out preference accuracy is "copied this rater," not "aligned."
- Not magic if the pairs are noisy. Two wrong answers still define an order.
- Offline: cannot invent replies the SFT model never wrote. PPO can sample new y .
- Absolute log-prob can fall on both sequences; if you only watch that, you will think training failed.

When not to. If you need on-policy exploration of completions that are not in the file, stay with RLHF. If you have no SFT reference, you do not have DPO's leash.

7. Teaching this note

About **35–40 minutes** at the board: write implicit r , plug into σ , compute $\sigma(0) = 1/2$. Play the Deep Dive **preference / RLHF** segment **2:48:26–3:09:39**. Lab 9 section 2 is the toy DPO scalar.

8. Worked example

Let $\beta = 1$. Suppose at init $\pi_\theta = \pi_{\text{ref}}$. Then

$$\log \frac{\pi_\theta(y_w)}{\pi_{\text{ref}}(y_w)} - \log \frac{\pi_\theta(y_l)}{\pi_{\text{ref}}(y_l)} = 0,$$

so $\mathcal{L} = -\log\sigma(0) = -\log(1/2) = \log 2$.

After a few steps, toy log-probs (sums over tokens): $\log\pi_\theta(y_w) = -2.0$, $\log\pi_{\text{ref}}(y_w) = -2.5$, $\log\pi_\theta(y_l) = -3.0$, $\log\pi_{\text{ref}}(y_l) = -2.0$. Log-ratio winner: $-2.0 - (-2.5) = 0.5$. Log-ratio loser: $-3.0 - (-2.0) = -1.0$. Difference $0.5 - (-1.0) = 1.5$. $\sigma(1.5) \approx 0.818$, loss $-\log 0.818 \approx 0.201$.

The frozen reference is still there: without it, π_θ can raise both log-probs or collapse. PPO can **sample new** y that were not in the pair file; offline DPO cannot.

9. Where students get stuck

- Forgetting $\sigma(0) = 1/2$ when $\pi_\theta = \pi_{\text{ref}}$ at step 0.
 - Dropping π_{ref} because “there is no RM.” The reference **is** the leash.
 - Expecting DPO to invent replies the SFT model never wrote.
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10. Video

[Karpathy — Deep Dive into LLMs like ChatGPT](#). Play the **preference** segment **2:48:26–3:09:39** (RLHF). Pause on comparison data: DPO uses those same pairs, without the PPO loop. The closed-form loss is the board.

11. Practice

1. If $\pi_\theta = \pi_{\text{ref}}$, what is $\mathcal{L}_{\text{DPO}}$? (Compute $\sigma(0)$.)
2. Why do you still keep a frozen reference if there is no RM?
3. Name one thing PPO can do that offline DPO cannot, given a fixed pair file.
4. $\beta = 2$, winner log-ratio 0.4, loser log-ratio 0.1. What is the argument of σ , and is the loss smaller or larger than $\log 2$?
5. Both completions have log-prob -10 under π_θ and under π_{ref} . What is the DPO loss, and what did the policy fail to learn yet?